

Evaluation of Research Stay Report

Toward Quantum Competitiveness in Adiabatic Quantum Computing: A Comparative Study of Variational Algorithms

Detailed assessment and paper-development remarks

Purpose	Evaluation of the research-stay report as a basis for a scientific paper.
Assessment	Strong research-stay report; major revision required before journal submission.

Overall assessment

Yes, this is a strong basis for a paper, but it is not yet a paper-ready manuscript. As a Master 2 research-stay report, I would rate it very good to excellent, roughly 8.5–9/10. It contains substantially more original computational work than a typical internship report: a unified RBM–VMC implementation, stochastic reconfiguration, several classical and hardware samplers, effective-temperature estimation, two generations of D-Wave hardware, an FPGA comparison, embedding studies, and an extension to frustrated Heisenberg models. The report also makes a commendable effort to distinguish observed competitiveness from quantum advantage.

For publication, however, it needs major revision. The central benchmark currently has a headline-level internal contradiction, the time-to- ϵ statistic is not yet rigorous enough, and the broad novelty claim overlaps strongly with earlier and very recent literature. My assessment is based on the report and its figures; I have not rerun the repository or audited the raw result caches.

Dimension	Assessment
Research-stay achievement	Excellent
Technical breadth	Very strong
Experimental maturity	Good, but uneven across subsections
Statistical rigor of the main comparison	Insufficient at present
Novelty positioning	Promising but not yet isolated
Paper readiness	Approximately halfway to a solid journal manuscript
Recommended decision	Accept as a strong report after corrections; do not submit as a paper yet

What is already strong

The report has four particularly valuable ingredients.

First, it establishes a reasonably coherent experimental framework instead of presenting disconnected solver benchmarks. The RBM ansatz, local-energy evaluation, stochastic reconfiguration, and sampler interface are integrated into one training pipeline. That gives the work more scientific value than a direct comparison of raw Ising-machine runtimes.

Second, the pooled CEM construction is potentially the most publishable methodological contribution. Instead of performing additional conditional sampling for fixed visible configurations, the report reuses the joint (v, h) samples produced during training and estimates a common β_{eff} from all visible-conditioned hidden observations. That is computationally attractive and partition-function free.

Third, the report includes useful negative results rather than hiding them. The hardware-native sparse RBM loses representational quality; parallel embeddings appear promising but are explicitly reported as statistically underpowered; and the positive-amplitude ansatz fails in the strongly frustrated regime.

Fourth, the reproducibility foundation is good: hardware parameters, seeds, solver settings, and a public code/data repository are provided.

The first issue to fix: the headline result is inconsistent

This is not a cosmetic problem. The abstract says that both Pegasus and Zephyr overtake FPGA at the largest system size.

But the detailed results say that at $N=64$, FPGA has $TTE_{99}=0.87$ s and is “clearly higher than Pegasus”; they do not say it is higher than Zephyr.

More importantly, Figure 15 on page 20 visibly places:

- Pegasus below FPGA at $N=64$;
- Zephyr at approximately 1.1–1.2 s;
- FPGA at approximately 0.87 s.

Thus, only Pegasus overtakes FPGA in the plotted metric. Nevertheless, the conclusion again says that both QPUs overtake FPGA.

The defensible current statement is closer to:

Under the device-time metric and the selected $h=0.5$, $\epsilon=0.1$ convergence criterion, Pegasus+CEM has a lower estimated TTE_{99} than FPGA at $N=64$. Zephyr+CEM does not. The result is a single observed crossover, not evidence of asymptotic scaling or quantum advantage.

That correction must be made in the abstract, results, Figure 15 discussion, and conclusion before the report is formally accepted.

Main scientific issues before publication

1. The present TTE_{99} definition underestimates retry cost

The report defines:

- p as the fraction of 20 seeds that validate;
- T_r as the median convergence time among the successful seeds;
- $TTE_{99}=T_r \log(0.01)/\log(1-p)$.

Failures are treated as runs that must be restarted.

The difficulty is that the standard restart formula assumes that each attempt has a fixed runtime. Here:

- a successful attempt can stop at t^* ;
- an unsuccessful attempt consumes the entire 100-iteration budget;
- T_r includes only successful, generally shorter, attempts.

Consequently, multiplying the median successful time by a retry factor omits the cost of failed full-budget runs. This matters precisely at $N=64$, where the ranking is affected by one to four unsuccessful seeds. The reported counts are FPGA 19/20, Zephyr 19/20, SA 18/20, Metropolis 17/20, and Gibbs 16/20.

There is also substantial uncertainty in those success probabilities. For orientation, exact 95% binomial intervals are approximately:

- 20/20: $p \in [0.832, 1]$;
- 19/20: $p \in [0.751, 0.999]$.

Therefore, treating 20/20 as $p=1$ with no retry penalty, while applying a penalty to 19/20, creates a sharp distinction that the available data do not statistically support.

A sound replacement would be one of the following:

1. Choose a fixed cutoff τ , estimate $p_\tau = \Pr(T \leq \tau)$, and calculate $T_{99}(\tau) = \tau \log(0.01)/\log(1-p_\tau)$. Report the minimum over prespecified cutoffs, with bootstrap confidence intervals.
2. Treat t^* as a right-censored time-to-event variable and use Kaplan–Meier or another survival estimator.
3. Simulate the complete retry process from the empirical distribution, including successful stopping times and full-budget failures.

The figure caption also reports a “median $T_{E_{99}}$ with IQR,” although the equations define one cell-level $T_{E_{99}}$ value from p and the median T_r . The origin of the plotted IQR therefore needs to be explicitly defined.

2. “Validation” is not presently independent

As written, convergence is determined from the rolling mean of the energy logged during training. No separate evaluator or held-out sample stream is described.

This is a serious problem for a sampler comparison. The local energy is an unbiased estimator of the variational energy only when samples are drawn from the intended $|\Psi_\theta|^2$ distribution. If one sampler returns a distorted distribution, it can produce:

- a biased SR update;
- a biased estimate of its own training energy;
- an apparently early threshold crossing.

Knowing the exact ground-state energy E_0 does not solve this: one must also estimate $E(\theta) = \langle \Psi_\theta | H | \Psi_\theta \rangle$ independently and accurately.

This is especially important because the QPU samples are visibly not exact Boltzmann samples. Even after choosing the best scaling in the small $N=8$ test, the report finds D_{TV} of approximately 30%, well above the finite-sample floor.

The paper should freeze each checkpointed ansatz and evaluate it using a common validation procedure:

- exact enumeration for approximately $N \leq 20-24$;
- long, independent, multi-chain reference sampling for larger N ;
- separate training and evaluation random streams;
- confidence intervals corrected for autocorrelation;
- preferably energy variance as a second criterion, because an exact eigenstate has zero variance.

A crossing should be declared only when an upper confidence bound on the independently estimated energy error falls below ϵ . Until then, “validated seed” should be renamed “seed satisfying the training-energy criterion.”

Your H100 servers make the independent offline evaluation comparatively inexpensive. The existing ansatz checkpoints can be reevaluated without additional QPU calls, provided they were archived.

3. The crossover is highly criterion dependent

The main benchmark uses only:

- the one-dimensional TFIM;
- $h=0.5$;
- $\epsilon=0.1$ per spin;
- four QPU sizes, $N \in \{8, 16, 32, 64\}$.

At $N=64$, $\epsilon=0.1$ per spin allows a total-energy error of 6.4. That may be acceptable as one coarse threshold, but it is too loose to support a general “competitiveness” conclusion by itself.

The report also shows that the target distribution changes dramatically with h : it is strongly peaked in the ordered regime and approaches a uniform distribution in the large-field regime. This means that a result at $h=0.5$ cannot be presumed representative of critical or paramagnetic sampling.

The paper needs time-to-accuracy curves, rather than one threshold:

$$\epsilon \in \{0.1, 0.05, 0.02, 0.01\},$$

at least for:

$$h \in \{0.5, 1.0, 2.0\}.$$

A stronger presentation would plot the lower envelope of wall time versus achieved independently evaluated error. Then readers can see whether the Pegasus crossover persists, disappears, or reverses when accuracy is tightened.

4. The reported power laws are not credible scaling evidence

The fits $\propto N^{0.16}$ and $\propto N^{0.05}$ use only four QPU sizes. Furthermore, each QPU iteration uses a fixed 20 μ s annealing time and a fixed number of reads.

A near-constant device access time under this protocol is not surprising. It does not imply that the underlying problem scales nearly constantly, because the dense $K_{N,N}$ RBM consumes increasing hardware resources through minor embedding.

The paper should avoid interpreting those four-point exponents as algorithmic scaling. At minimum it should report, for every N :

- number of physical qubits used;
- mean and maximum chain length;
- chain-break fraction;
- coefficient-rescaling range;
- embedding success rate;
- number of simultaneously available embeddings;
- QPU programming, annealing, and readout components separately.

A more honest timing model would be

$$t(N) = t_0 + aN^p,$$

where t_0 captures the large fixed call overhead. With only four points, even that fit should be descriptive rather than inferential.

5. The timing boundaries are not yet apples-to-apples

The comparison currently uses:

- D-Wave qpu_access_time, excluding network latency and one-time embedding;
- classical sampler kernel time after excluding JAX compilation;

- FPGA on-device cycle count divided by a 120 MHz clock.

This is a legitimate device-kernel benchmark, but it is not yet an end-to-end time-to-solution benchmark.

The paper should provide two clearly labelled results:

1. Device/kernel time, approximately the current metric.
2. End-to-end wall time, including data preparation, host-device transfer, embedding/unembedding, chain decoding, CEM, and network calls.

A third, practically useful metric would be amortized end-to-end time when an embedding is cached and many training jobs are batched.

The report deliberately excludes SR because it is nominally common across solvers. That is defensible for isolating sampler performance, but total training time should still be shown. Otherwise a large sampler speedup might be irrelevant once the SR solve dominates.

6. The FPGA baseline is insufficiently documented

The FPGA is central to the abstract and conclusion, yet the report provides almost no algorithmic description of it. The appendix gives Float32 precision, a 120 MHz clock, 1024 replicas, and a cycle-count timing convention, but not enough information to reproduce or scientifically interpret the sampler.

A paper needs to state:

- the exact FPGA algorithm and transition/update equations;
- whether it has a known stationary distribution;
- how its effective temperature is estimated or controlled;
- how 1024 replicas are converted into the 200 samples used by SR;
- whether samples are independent;
- number of update steps and parameter choices;
- FPGA model, toolchain, bitstream version, memory traffic, and host-transfer costs;
- sampling-fidelity results analogous to Figure 5;
- whether solver parameters were tuned per instance.

Without this, “FPGA is the fastest and most energy-efficient classical solver” is difficult to assess.

The energy conclusion is even less mature. QPU energy is correctly omitted, but FPGA energy is based on an assumed constant 45 W, while the report does not describe the GPU power-integration procedure or common system boundaries. The energy panel should either be moved to supplementary material and labelled exploratory or rebuilt using measured board energy, idle subtraction, polling uncertainty, and consistent inclusion of host overhead.

7. The classical baselines need an optimization-budget comparison

The report describes the benchmark as hyperparameter matched, but the inner sampling workloads differ substantially:

- Metropolis: fresh chains, 200 warm-up sweeps plus one sweep;
- Gibbs: persistent chains, ten block sweeps after a one-time initialization;
- SA: fresh chains, 200 warm-up sweeps plus 40 cooling sweeps.

Matching outer SR parameters is scientifically useful, but it is not equivalent to comparing the best achievable time-to-accuracy of each sampler. In particular, the report's earlier sampling-quality figures show that SA quality can continue improving far beyond 40 sweeps.

A convincing paper should report both:

- fixed-protocol comparison, preserving the present controlled experiment;
- solver-optimized comparison, where every sampler receives the same hyperparameter-tuning budget and minimizes independently evaluated time-to-accuracy.

Given the available fast GPU/FPGA simulated-bifurcation implementation, an LSB/SBM-type sampler would also be a natural strong baseline. It should, however, be treated as a sampler and calibrated against a target distribution—not merely judged by its ability to find low energies.

Novelty relative to current work

The broad claim “D-Wave-assisted RBM variational Monte Carlo for the TFIM” cannot be the novelty. A 2018 Physical Review B paper already used an RBM wavefunction, VMC, and a D-Wave sampler for the TFIM; that work reached $L=64$ on the 2000Q hardware.

(<https://link.aps.org/doi/10.1103/PhysRevB.98.184304>)

Temperature calibration alone is also a crowded claim. A 2025 online-calibration paper already used returned visible and hidden samples to estimate internal quantum-annealer parameters during RBM training, including models with more than one calibration parameter.

(<https://arxiv.org/abs/2307.09785v2>) The recent CEM work estimates effective inverse temperature from conditional hidden expectations at a fixed visible condition.

(<https://arxiv.org/html/2512.02323v1>) In addition, a 2026 Physical Review E article reports temperature-controlled quantum-annealing samples for RBM training and claims faster convergence and lower validation error than classical sampling.

(<https://link.aps.org/doi/10.1103/2g6m-whm2>)

This does not mean the present project lacks a paper. It means the novelty must be much more precise.

The strongest paper hidden inside this report

I would make the paper about:

Partition-function-free online conditional thermometry for quantum-annealer-assisted neural quantum states, evaluated by independently validated time-to-accuracy and hardware-resource accounting.

The distinct methodological point is that every observed visible configuration v_i is treated as its own conditioning vector, and all paired hidden observations are pooled. This avoids both exact partition functions and additional fixed-condition queries.

The present least-squares objective could be strengthened further. For

$$\Theta_{ij} = b_j + \sum_k v_{ik} W_{kj}$$

the conditional likelihood under a common inverse temperature is

$$\ell(\beta) = \sum_{i,j} [\beta h_{ij} \Theta_{ij} - \log(2 \cosh(\beta \Theta_{ij}))].$$

This produces the score equation

$$\sum_{i,j} \Theta_{ij} [h_{ij} - \tanh(\beta \Theta_{ij})] = 0.$$

That gives a natural conditional maximum-likelihood or pseudolikelihood estimator. It would be valuable to:

- compare it with the current pooled least-squares CEM;
- prove identifiability and consistency under the ideal Gibbs model;
- derive Fisher information or asymptotic variance;
- use a block bootstrap or sandwich estimate for correlated QPU reads;
- quantify misspecification when one scalar β is insufficient;
- derive a bound connecting $|\beta - \hat{\beta}|$ to bias in the SR force vector.

That theoretical layer could turn a competent benchmarking paper into a considerably stronger methods paper.

Recommended paper structure

The main paper should contain:

1. RBM-VMC problem and clearly enumerated contributions.
2. Pooled conditional thermometry and its statistical properties.
3. Independent validation and benchmark methodology.
4. Calibration accuracy and residual non-Gibbs distortion.
5. Time-to-accuracy over N , h , and ϵ .
6. Device time, end-to-end time, and embedding-resource analysis.
7. Limitations and conditions under which a crossover is observed.

Move the following out of the central narrative:

- introductory textbook descriptions of Metropolis, Gibbs, JAX, and the TFIM solution;
- full convergence grids;
- most of the Marshall-sign discussion;
- the underpowered parallel-embedding result;
- the exploratory energy comparison.

The sparse-RBM result can remain as a short resource-accuracy section, but the conclusion should be narrowed. The current experiment shows that the tested hardware-derived masks are poor for the tested RBM/TFIM configurations; it does not establish that all sparse or hardware-native neural quantum states are unsuitable.

A more ambitious follow-up would construct physics-aware local or convolutional RBMs rather than deleting edges from a dense RBM according to the hardware graph.

Better titles

The present title is misleading: only one variational algorithm is studied, while multiple samplers are compared. “Adiabatic quantum computing” is also not the most accurate emphasis for finite-temperature sampling on an open-system quantum annealer.

Stronger options are:

- Online Thermometry for Quantum-Annealer-Assisted Variational Monte Carlo
- Benchmarking Temperature-Calibrated Quantum Annealing as a Sampler for Neural Quantum States

- Time-to-Accuracy Benchmarking of Quantum Annealers in Neural-Network Variational Monte Carlo
- Pooled Conditional Thermometry for Quantum-Annealing Boltzmann Samplers

A practical minimum revision plan

The existing work can be converted into a paper without rerunning every experiment from scratch.

1. Correct the Pegasus/Zephyr claim immediately. Recompute Figure 15 directly from the archived raw cells and verify all labels, retry multipliers, and units.
2. Reanalyse the stored trajectories. Produce accuracy-versus-time curves for several ϵ values and bootstrap them at the seed level. This requires no new QPU calls.
3. Independently reevaluate archived checkpoints. Use exact enumeration at small N and long reference sampling on the H100s for larger N . This is the most important methodological repair.
4. Concentrate new QPU budget near the claimed crossover. Use $N=32,48,64$, at least $h=0.5$ and $h=1$, and approximately 50–100 seeds at $N=64$. A sequential stopping rule can avoid wasting calls if the ranking becomes clear.
5. Add hardware controls. Use several spin-reversal gauges, multiple embeddings, at least a small chain-strength and anneal-time sweep, and report chain-break statistics and physical-qubit counts.
6. Document and calibrate the FPGA solver completely. Add sampling-fidelity experiments and distinguish on-device, host-device, and end-to-end time.
7. Compare thermometry methods. At minimum: exact KL-based β at small N ; original fixed-condition CEM; current pooled CEM; conditional maximum likelihood; one- versus multi-parameter calibration.

Without additional QPU experiments, a publishable but narrower methods paper may still be possible around pooled thermometry. The “quantum competitiveness” and scaling claims would then need to be substantially softened.

Technical and editorial corrections

- On page 17, the TFIM transverse-field contribution is written as $\sigma_i^x \sigma_j^x$; Equation 2.1 contains the single-spin term σ_i^x .
- The quantity called “sparsity rate” is defined as selected connections divided by all possible connections. That is a connectivity density; sparsity would normally be one minus that quantity.
- Replace “Sorello” with “Sorella.”
- Use “effective inverse temperature” consistently for β_{eff} .
- State the bounds used in every scalar minimization and report how often a bound is hit.
- Figure 9 should report numerical RMSE, relative error, bias, and boundary-hit frequency rather than saying only that CEM is “very close” to the exact estimate. The figure and caption already show that some points are problematic.
- Figure 16 is too dense for journal dimensions. Split it into model-specific figures or move the full grid to supplementary material.
- The Figure 15 float currently splits the first sentence of the conclusion across pages 19 and 20.
- “Median GPU energy” is not an appropriate label for a panel that includes an FPGA estimate.

- Avoid saying that the QPU “targets p_θ ” without qualification; it is intended to approximate p_θ , and the residual goodness-of-fit must be measured.
- Power-law exponents from four QPU points should either be removed or accompanied by uncertainty intervals and a strong warning that they are descriptive only.

Final recommendation

As a research-stay report: this is excellent work and demonstrates genuine research independence. I would recommend a highly positive evaluation after the false “both QPUs overtake FPGA” statement and the most obvious technical errors are corrected.

As a paper: the material is clearly sufficient, but the current manuscript tries to combine too many stories:

- quantum-annealer thermometry;
- solver time/energy benchmarking;
- sparse hardware-native RBMs;
- parallel embeddings;
- the Marshall sign rule.

The strongest publication is one focused paper on pooled thermometry plus independently validated time-to-accuracy. The sparse and parallel-embedding results can support the hardware-resource discussion or become a later paper after additional theory and data. The Marshall section is useful in the report but is not presently a central novel result.

After the statistical and methodological repairs, Physical Review Applied, Physical Review E, or Quantum Science and Technology would be realistic targets. Physical Review Research becomes plausible if the thermometry estimator receives a genuine theoretical analysis and the experiments are broadened. A submission to Quantum would require considerably stronger generality, independent validation, and more convincing evidence that the contribution extends beyond this particular RBM-TFIM-D-Wave configuration.